

## YARRAWONGA, Australia

WMO: 948620

Lat: 36.0294S

Lon: 146.0306E

Elev: 129

StdP: 99.78

Time Zone: 10.00 (AUS)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | -0.1       | 1.0 | -3.4                          | 2.9 | 20.9 | -1.8 | 3.3 | 14.9 | 10.2                  | 11.9 | 9.3 | 11.6 | 1.4                   | 240  | 0.504 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 16.0                   | 38.6            | 19.3 | 36.1 | 18.9 | 34.0 | 18.4 | 22.2                | 29.3 | 21.2 | 29.2 | 20.4 | 29.0 | 4.9                  | 330  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.6                            | 15.5 | 24.5 | 19.2 | 14.2 | 23.8 | 17.7 | 12.9 | 23.0 | 65.8          | 28.7 | 62.2 | 29.0 | 59.1 | 28.6 | 26.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.1              | 8.8  | 7.8 | -2.1                       | 42.2 | 1.2                | 1.5 | -3.0                                               | 43.3 | -3.7       | 44.2 | -4.3       | 45.1 | -5.2       | 46.2 |
|                   |      |     | -2.4                       | 23.7 | 1.2                | 1.4 | -3.2                                               | 24.8 | -3.9       | 25.6 | -4.6       | 26.4 | -5.5       | 27.4 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 15.9   | 24.6  | 23.5  | 20.7  | 16.3  | 12.0  | 9.1   | 8.6   | 9.2   | 11.4  | 15.1  | 19.3  | 21.9  |
|                                                                        | DBStd       | 6.53   | 4.14  | 3.75  | 3.70  | 3.17  | 2.54  | 2.22  | 2.05  | 2.22  | 2.76  | 3.58  | 4.10  | 3.92  |
|                                                                        | HDD10.0     | 162    | 0     | 0     | 0     | 1     | 8     | 42    | 53    | 43    | 14    | 1     | 0     | 0     |
|                                                                        | HDD18.3     | 1527   | 2     | 4     | 18    | 77    | 197   | 276   | 302   | 284   | 208   | 114   | 36    | 11    |
|                                                                        | CDD10.0     | 2326   | 452   | 379   | 331   | 189   | 70    | 16    | 9     | 17    | 56    | 159   | 279   | 367   |
|                                                                        | CDD18.3     | 649    | 196   | 149   | 91    | 15    | 0     | 0     | 0     | 0     | 1     | 13    | 64    | 120   |
|                                                                        | CDH23.3     | 8576   | 2685  | 1758  | 1041  | 174   | 3     | 0     | 0     | 0     | 23    | 284   | 966   | 1642  |
|                                                                        | CDH26.7     | 4202   | 1518  | 880   | 426   | 32    | 0     | 0     | 0     | 0     | 4     | 92    | 441   | 808   |
| (14) Wind                                                              | WSAvg       | 3.5    | 4.1   | 3.8   | 3.6   | 3.1   | 3.0   | 2.8   | 3.2   | 3.3   | 3.6   | 3.7   | 4.1   | 4.2   |
| (15) Precipitation                                                     | PrecAvg     | 539    | 39    | 30    | 32    | 41    | 57    | 44    | 54    | 59    | 50    | 49    | 38    | 44    |
|                                                                        | PrecMax     | 1068   | 158   | 163   | 115   | 146   | 159   | 121   | 126   | 133   | 118   | 188   | 125   | 157   |
|                                                                        | PrecMin     | 239    | 1     | 0     | 0     | 4     | 3     | 6     | 4     | 10    | 7     | 10    | 2     | 0     |
|                                                                        | PrecStd     | 188    | 36    | 40    | 28    | 33    | 39    | 29    | 32    | 30    | 28    | 43    | 27    | 39    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 42.1  | 40.7  | 36.5  | 29.9  | 22.8  | 17.9  | 17.0  | 19.9  | 26.4  | 33.2  | 37.0  | 39.9  |
|                                                                        |             | MCWB   | 20.1  | 21.2  | 19.0  | 17.2  | 14.3  | 12.7  | 12.6  | 13.2  | 14.1  | 16.1  | 18.0  | 18.7  |
|                                                                        | 2%          | DB     | 39.6  | 36.8  | 33.5  | 27.2  | 20.6  | 16.1  | 15.1  | 17.5  | 22.5  | 29.1  | 34.1  | 36.5  |
|                                                                        |             | MCWB   | 19.8  | 19.4  | 18.7  | 16.1  | 13.8  | 11.6  | 11.5  | 12.0  | 14.1  | 15.5  | 17.5  | 18.4  |
|                                                                        | 5%          | DB     | 36.8  | 34.1  | 31.1  | 25.3  | 18.9  | 14.9  | 13.8  | 15.7  | 20.1  | 26.5  | 31.6  | 33.7  |
|                                                                        |             | MCWB   | 19.6  | 19.1  | 18.2  | 15.5  | 13.3  | 11.2  | 10.9  | 11.1  | 12.9  | 15.1  | 16.9  | 17.7  |
|                                                                        | 10%         | DB     | 34.1  | 31.9  | 28.9  | 23.4  | 17.3  | 13.6  | 12.6  | 14.2  | 18.2  | 23.9  | 29.0  | 31.3  |
|                                                                        |             | MCWB   | 18.7  | 18.4  | 17.2  | 14.9  | 12.6  | 10.7  | 10.1  | 10.5  | 12.2  | 14.4  | 16.2  | 17.3  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 23.5  | 24.0  | 22.5  | 18.9  | 16.7  | 13.9  | 14.0  | 14.1  | 16.7  | 19.2  | 21.3  | 23.0  |
|                                                                        |             | MCDB   | 31.6  | 32.7  | 28.5  | 24.7  | 19.1  | 15.3  | 15.1  | 17.3  | 21.7  | 25.8  | 30.7  | 28.3  |
|                                                                        | 2%          | WB     | 22.1  | 22.1  | 20.7  | 17.7  | 15.1  | 12.8  | 12.2  | 12.8  | 15.1  | 17.6  | 19.7  | 21.0  |
|                                                                        |             | MCDB   | 31.0  | 28.0  | 27.9  | 23.5  | 18.4  | 14.6  | 14.0  | 16.3  | 20.0  | 24.3  | 27.0  | 27.2  |
|                                                                        | 5%          | WB     | 21.1  | 21.1  | 19.6  | 16.7  | 14.1  | 11.9  | 11.2  | 11.7  | 13.9  | 16.5  | 18.6  | 19.6  |
|                                                                        |             | MCDB   | 30.6  | 28.2  | 27.3  | 22.4  | 17.4  | 14.0  | 13.1  | 14.9  | 18.6  | 23.1  | 26.1  | 28.4  |
|                                                                        | 10%         | WB     | 20.3  | 20.0  | 18.6  | 15.8  | 13.1  | 11.1  | 10.3  | 10.8  | 12.7  | 15.3  | 17.6  | 18.6  |
|                                                                        |             | MCDB   | 30.7  | 28.4  | 26.5  | 21.8  | 16.5  | 13.1  | 12.3  | 13.8  | 17.2  | 22.3  | 25.5  | 28.3  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 16.0  | 14.6  | 14.0  | 12.9  | 11.0  | 9.6   | 8.6   | 10.6  | 13.0  | 15.1  | 15.5  | 15.5  |
|                                                                        |             | MCDBR  | 19.0  | 17.5  | 16.4  | 15.4  | 13.0  | 11.3  | 10.0  | 13.0  | 16.4  | 19.2  | 19.1  | 18.9  |
|                                                                        |             | MCWBR  | 5.4   | 5.6   | 5.7   | 6.4   | 6.9   | 7.2   | 6.6   | 8.1   | 8.6   | 8.0   | 6.2   | 5.8   |
|                                                                        | 5% WB       | MCDBR  | 14.8  | 12.7  | 13.1  | 12.3  | 10.4  | 9.3   | 8.6   | 11.1  | 14.1  | 15.0  | 14.5  | 13.8  |
|                                                                        |             | MCWBR  | 5.4   | 4.9   | 5.5   | 5.7   | 6.1   | 6.4   | 6.1   | 7.3   | 8.4   | 7.6   | 5.7   | 5.5   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.347 | 0.345 | 0.321 | 0.312 | 0.301 | 0.295 | 0.289 | 0.297 | 0.326 | 0.327 | 0.344 | 0.339 |
|                                                                        | taud        |        | 2.470 | 2.495 | 2.557 | 2.546 | 2.554 | 2.571 | 2.587 | 2.542 | 2.446 | 2.470 | 2.463 | 2.486 |
|                                                                        | Ebn at Noon |        | 987   | 967   | 954   | 903   | 857   | 835   | 864   | 907   | 932   | 974   | 985   | 999   |
|                                                                        | Edh at Noon |        | 117   | 109   | 95    | 85    | 73    | 67    | 70    | 83    | 104   | 111   | 117   | 116   |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 7.73  | 6.79  | 5.64  | 4.06  | 2.81  | 2.16  | 2.34  | 3.26  | 4.62  | 6.06  | 7.00  | 7.75  |
|                                                                        | RadStd      |        | 0.43  | 0.39  | 0.28  | 0.19  | 0.18  | 0.23  | 0.13  | 0.25  | 0.41  | 0.48  | 0.37  | 0.32  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (25 neighbors) | +0.44 | N/A     | N/A    | +1.05   | N/A   | N/A   | N/A         | N/A     | +139    | +90     |

Nomenclature: See separate page